

Upwire: Delivering customized messaging services for businesses with Google Cloud

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[About Upwire](#)



Using Google Cloud, Upwire allowed businesses to deliver fast, reliable messaging services while fostering innovation internally and at their customers' businesses. The company is now using the flexibility and scalability of Google Cloud Kubernetes Engine to scale its services.

About Upwire

Founded in 2013 and headquartered in Sydney, Upwire provides communication workflows to business customers worldwide. The business has a presence in Australia, Europe, and the United States and aims to deliver meaningful solutions for customers through its drag-and-drop interface that allows them to create communication solutions that deliver better outcomes. This interface allows Upwire's customers to create interactive voice response (IVR) systems, two-factor authentication, SMS chatbots for customer service, two-way SMS

customized business
messaging service
customers.

campaigns, SMS and IVR credit card payment
services, and voice biometric authentication.

Industries: Technology

Location: Australia

Google Cloud results

- Helps power a 10x increase in revenue over the past three years
- Cuts proof-of-concept development costs from more than \$100K to less than \$100
- Delivers near-100% availability and enables business to meet message delivery SLAs

Process
server,
voice,
email
per m

enterprise/)

Kubernetes Engine

Google Cloud (<https://cloud.google.com/>)

Dialogflow Enterprise Edition

(<https://cloud.google.com/dialogflow-enterprise/>)

Engine (<https://cloud.google.com/appengine/>),

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Storage (<https://cloud.google.com/storage/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

SQL (<https://cloud.google.com/sql/>)

App Engine (<https://cloud.google.com/appengine/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Text to Speech

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud Datastore

(<https://cloud.google.com/datastore/>)

Cloud SQL (<https://cloud.google.com/sql/>) help.

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Speech to Text

(<https://cloud.google.com/speech/>)

Cloud Text to Speech

(<https://cloud.google.com/text-to-speech/>)

- Empowers
people
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solutions

Headquartered in Sydney, Upwire (<https://www.upwire.com/>) provides business messaging workflows to customers worldwide from offices in Australia, Europe, and the United States. The business provides drag-and-drop interfaces that enable businesses to create interactive voice response (IVR) systems, two-factor authentication, SMS chatbots for customer service, two-way SMS campaigns, SMS and IVR credit card payment services, and voice biometric authentication.

Upwire describes its workflows as being built for people with no IT skills at all—just drag and click each step of the way. “We deliver outcomes for our customers through our drag-and-drop interface that allows them to create communication solutions,” says Ben Brophy, co-founder of Upwire.

Upwire is the culmination of a journey that started in 2002. Ben says he and fellow co-founder, Shane Berkinshaw, began developing the concept that became Upwire when working for an established communications business.

“At that time, industry participants relied on traditional infrastructures to run their businesses,” Ben explains. “Business messaging providers had to close deals with customers that required considerable investment in infrastructure. The costs involved were such that providers faced an uphill task adding more infrastructure for redundancy, and this increased the risk to services if a problem occurred.”

Providers also faced the problem of not being able to plan for peaks and troughs in customer usage yet having to operate an infrastructure that enabled one customer to send 10 million messages without impacting another customer’s services.

The emergence of public cloud services prompted Ben and Shane in 2008 to create a business that ran infrastructure and associated products in the cloud. “We were able to do a lot of heavy lifting for large corporations that needed assistance with activities such as database cleansing,” says Ben. “Companies could connect and deliver messages to our products and infrastructure via APIs.

However, the founders wanted to unleash the creativity and innovation they believed was limited to people with coding skills. “In 2013—after the release of HTML5 allowed us to effectively create drag-and-drop environments—we mapped out a solution we thought would enable anyone in the world to be truly innovative,” says Ben. “It would allow people without coding skills to create high-quality communications systems, and more than that, it would allow true democratization of technology.”

Rapidly expanding

Ben and Shane then created Upwire, a business that is expanding quickly and now has about 14 team members globally. “We have offices in the United Kingdom, the United States, and Australia and clients all over the world,” says Shane. “Upwire is resonating in all regions in which we operate because of our outcomes-oriented approach and the fact we can deliver our solutions in days, whereas traditional products such as customer-relationship management systems and enterprise-resource planning systems can take 18 months or longer to implement and run in production. Businesses realize they cannot take that time to move in the current climate.”

Starting the new business in an environment of increasingly mature cloud services prompted the founders to move from a multicloud architecture to a single cloud platform. Having established close relationships with product managers and developers of products such as [App Engine](#) (/appengine), Ben and Shane understood enough about [Google Cloud](#) (<https://cloud.google.com/>) to realize the platform was the best fit for Upwire.

“We were going to take on large enterprise and corporate customers and we needed to be comfortable with a service provider’s capabilities in encryption, redundancy, and backups,” says Ben. “We also needed to be sure Google’s product portfolio and investments aligned with our need to continually innovate.”

Creating Upwire Playground

The fact Google hosts customers on the same infrastructure that it runs its own business on also prompted the founders to consolidate onto Google Cloud. Upwire initially used App Engine to run applications in an infrastructure-abstracted environment before expanding to products such as [Compute Engine](#) (/compute) for infrastructure as a service, [Cloud Storage](#) (/storage) for unified object storage, [Datastore](#) (/datastore) to run a NoSQL database, and [Cloud SQL](#) (/sql) to run its relational databases.

Machine learning-powered [Dialogflow](#) (/dialogflow) powers interfaces with Upwire modules that in turn integrate with third parties to run customer service chatbots. In addition, Upwire has integrated [Text-to-Speech](#) (/text-to-speech) and [Speech-to-Text](#) (/speech-to-text) and made them configurable objects on its platform to enhance its offering to customers. Google Translate powers the translation of Upwire-powered services into a wide range of languages.

Upwire is now using the Google Kubernetes Engine service designed to run containerized applications to rapidly deploy new products such as Upwire Playground, which allows the business to reuse and redeploy its technologies to meet customer-specific requirements. “If a customer says to us we want to run our own isolated and white-labelled version of Upwire, we simply deploy a new project in Kubernetes and set them up with their own Upwire Playground,” says Ben. Over time, Upwire also plans

to transition its messaging engine from App Engine into the Kubernetes environment.

Upwire Playground will enable both the business and customers to build modules and write code to, for example, query a database as to whether a person has made a payment and, if they have failed to do so, initiate the required workflows. “Previously, we had to use Cloud Functions—which provides scalable ‘functions as a service’—to shift data into the right format or analyze data to conduct the advanced evaluations increasingly demanded by customers,” says Ben. “Using the payment example, a customer may want to know how many days overdue the individual is.

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—Ben Brophy, co-founder,
Upwire

Proofs of concept at minimal cost

Running on Google Cloud enables Upwire to complete proofs of concept at minimal cost in just a few days. “Customers were used to proofs of concept (PoC) that took weeks and cost \$100,000, including the cost of setting up servers,” explains Ben. “With Google Cloud, we can complete some PoCs in days for under \$100—needless to say, customers have been very surprised. In addition, Upwire running on Google Cloud helps businesses relieve pressure on scarce coding resources and allows people who understand business logic to innovate.”

Google Cloud’s pricing structure enables the business to align infrastructure costs with growth. This allows the business to cost-effectively expand into new markets without the burden of considerable capital expenditure on infrastructure. At the same time, Upwire can dedicate the savings to building a better product for customers.

To date, Upwire has gained the confidence of near-100% availability with global reach by running its product on Google Cloud in multiple regions worldwide. “Even if something went wrong at an infrastructure level, we and our customers have the considerable engineering resources of Google Cloud working to fix it within minutes,” says Ben.

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—Ben Brophy, co-founder,
Upwire

Handling huge volumes of messages

Perhaps most important to Upwire is how Google Cloud can reliably deliver large volumes of business messages quickly. “We deliver millions of messages ranging from password resets to trade confirmations in the United States, and they are all time critical,” says Ben. “We have to meet a range of service level agreements.” For example, the company might have to deliver time-stamped emails within a set number of seconds and show that they have been delivered to a domain or a telecommunications provider.

Ben points out that Upwire’s business is considerably more complex than simply moving messages back and forth. “Our product handles the movement of rich content such as micro-web forms that capture data,” he explains. For instance, a system might help manage registrations by sending an automated SMS to people who have not sent in all the required documentation. The SMS includes a link that people can click to access a traditional content microform that can be used to capture a driver’s license or other missing information. It is then encrypted and sent back to the customer.

“With support from Google Cloud, Upwire easily handles all the messaging and business logic associated with the various threads and interactions this service generates,” says Ben. Overall, Google Cloud has to support millions of voice minutes, SMS messages, and emails every month.

Over the last three years, Upwire has experienced a 10x increase in revenue and messaging volumes have increased sharply—messaging through its voice product alone has grown by several thousand percent. The business is also negotiating two global partnerships expected to deliver another dramatic increase in volumes.

Data warehouse supports analytics

The business also uses a BigQuery data analytics warehouse to rate and analyze message delivery, with the focus on performance into several countries. The warehouse supports internal and customer reporting and an alert system triggered when certain thresholds are breached.

Overall, Shane says, Google Cloud enables the business to focus on developing powerful tools for customers to communicate and deliver outcomes, rather than on infrastructure management. “Our business literally would not exist without the flexibility, agility, and scalability provided by Google Cloud,” he says.